

NON-NEGATIVE MATRIX FACTORIZATION

AP

REVIEW OF SPECTRAL ANALYSIS/SVD

FROM DECOMPOSITION TO DATA COMPREHENSION

Decompose the data matrix and interpret its 'top' eigenpairs as concepts/topics for user and activity classification:

$$M = U\Sigma V^T$$

$U_{(m \times r)}$ is column-orthonormal: if $\mathbf{u}_i$ and $\mathbf{u}_j$ are columns of U , then $\mathbf{u}_i^T \cdot \mathbf{u}_j = 0$

$V_{(r \times n)}^T$ is row-orthonormal: $\mathbf{v}_i^T \cdot \mathbf{v}_j = 0$

$\Sigma_{(r \times r)}$ (or $D_{(r \times r)}$) is diagonal, with the σ_i s on the main diagonal.

dimension $r \leq \min\{m, n\}$ also depends on the no. of singular values found; reduce it further by dropping 0/very low σ s

HURDLE: INTERPRETATION OF NEGATIVE VALUES

■ $A = U \Sigma V^T$ - example:

Diagram illustrating the SVD decomposition $A = U \Sigma V^T$ for a movie rating matrix A .

Matrix A (Movie Ratings):

	Matrix	Alien	Serenity	Casablanca	Amelie
SciFi	1	1	1	0	0
	3	3	3	0	0
	4	4	4	0	0
	5	5	5	0	0
Romnce	0	2	0	4	4
	0	0	0	5	5
	0	1	0	2	2

Matrix U (Left Singular Vectors):

0.13	0.02	-0.01
0.41	0.07	-0.03
0.55	0.09	-0.04
0.68	0.11	-0.05
0.15	-0.59	0.65
0.07	-0.73	-0.67
0.07	-0.29	0.32

Matrix Σ (Singular Values):

12.4	0	0
0	9.5	0
0	0	1.3

Matrix V^T (Right Singular Vectors):

0.56	0.59	0.56	0.09	0.09
0.12	-0.02	0.12	-0.69	-0.69
0.40	-0.80	0.40	0.09	0.09

Interpretation:

- The **SciFi-concept** is represented by the first column of U .
- The **"strength" of the SciFi-concept** is represented by the first singular value, **12.4**.
- The **Romnce** concept is represented by the first column of V^T .

With negative values from SVD we cannot distinguish between

- i. lack of information,
- ii. lack of interest or
- ii. outright *repulsion*.

AN IDEAL VIEW

A non-negative decomposition of the activity matrix would be *interpretable*:

$$A_{(m \times n)} = P_{(m \times r)} \cdot Q_{(r \times n)}$$

- A : activity
- P : user participation to a topic
- Q : quantity of the topic in product

user/product profiling and recommender sys. would be half-done already!

P and Q are of independent interest

NON-NEGATIVE DECOMPOSITION

CONTEXT

Lee and Seung sought a new approach to the problem of learning images (how to recognise them) via visual patterns:

reduce a specific image to the combination of known patterns: face $\approx$ eyes, eyebrows, nose, chin etc.

Could each face be decomposed into basic, repeated visual patterns?

In the neural network model of brain activity, firing rates of neurons cannot be negative: patterns are either activated or ignored

Sparsity: only few patterns should be needed to re-create an image

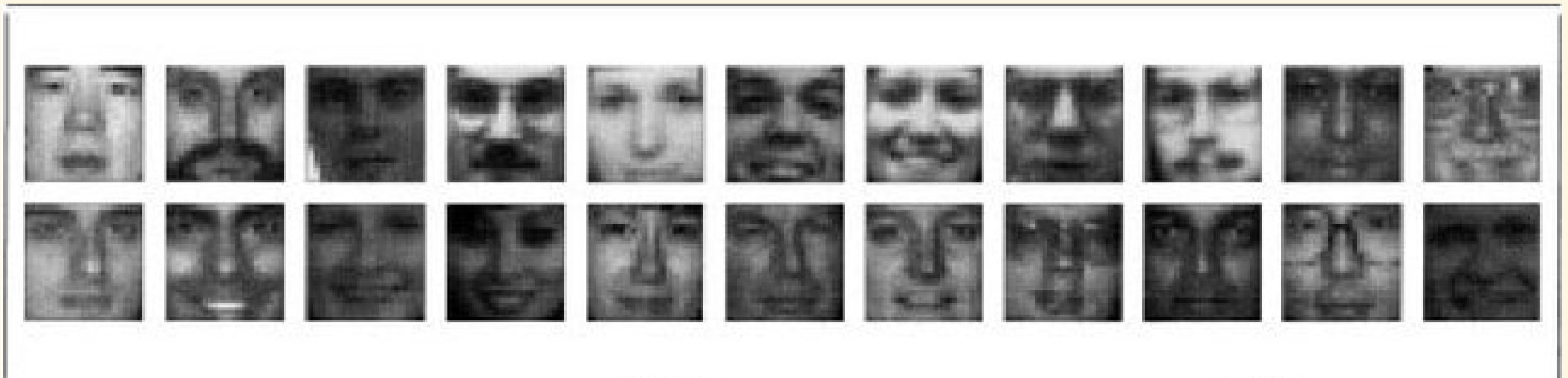
Example dataset: faces from 'the wild' [Papageorgiou-Poggio] (not Olivetti)

2429 mugshots, each having $19 \times 19 = 361$ pixels, rescaled to $[0,1]$:

$$V_{361 \times 2429}$$

Lee-Seung fix $r = 49$ and compute $V_{361 \times 2429} = W_{361 \times 49} \times H_{49 \times 2429}$

For comparison, they also try PCA (diagonalisation of VV^T) and VQ



THE NUMERICAL PROBLEM

Instance: a non-negative matrix V

Solution: non-negative matrix factors W and H s.t.

$$V \approx W \cdot H$$

with $w_{ij}, h_{rs} \geq 0$

EXAMPLE

$$V_{3 \times 3} = \begin{pmatrix} 4 & 0 & 2 \\ 0 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

choose $r = 2$

$$V_{3 \times 3} \approx W_{3 \times 2} \times H_{2 \times 3} = \begin{pmatrix} 2 & 0 \\ 0 & 1.5 \\ 1 & 1 \end{pmatrix} \times \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 0 & 2 \\ 0 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix} \approx \begin{pmatrix} 4 & 0 & 2 \\ 0 & 3 & 1.5 \\ 2 & 2 & 2 \end{pmatrix}$$

NOTATION

$$A = B \cdot C$$

Let $\mathbf{a}_i$ be the i -th column of A . It can be expressed as

$$\mathbf{a}_i = B \cdot \mathbf{c}_i$$

each col. of the result is seen as a linear combination of the cols. of B , with $\mathbf{c}_i$ supplying the *weights*:

$$\mathbf{a}_i = B \cdot \mathbf{c}_i = c_{1,i} \mathbf{b}_1 + c_{2,i} \mathbf{b}_2 + \cdots + c_{n,i} \mathbf{b}_n$$

EXAMPLE

$$\begin{pmatrix} 4 & 0 & 2 \\ 0 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix} \approx \begin{pmatrix} 2 & 0 \\ 0 & 1.5 \\ 1 & 1 \end{pmatrix} \times \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} \approx \begin{pmatrix} 2 & 0 \\ 0 & 1.5 \\ 1 & 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} \approx \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \cdot 2 + \begin{pmatrix} 0 \\ 1.5 \\ 1 \end{pmatrix} \cdot 0$$

INTERPRETABILITY OF NMF

HOW WE READ NMF

Let $\mathbf{v}_i$ be the i -th column of V .

If V is an activity m., $\mathbf{v}_i$ represent the *consumption* of i

$$\mathbf{v}_i \approx W \cdot h_i$$

Consumption of i is given by a linear combination of the cols. of W , with h_i supplying the weights.

Each $\mathbf{w}_i$ is interpretable as a pattern (or mask)

[Lee & Seung, Nature, 1999]: “Learning the parts of objects by non-negative matrix factorization.”

$$\mathbf{v}_i \approx \mathbf{w}_1 \cdot h_{1,i} + \dots \mathbf{w}_r \cdot h_{r,i}$$

W can be regarded as containing a basis that is optimized for the linear approximation of the data in V.

Since relatively few basis vectors are used to represent many data vectors, good approximation can only be achieved if the basis vectors discover structure that is latent in the data.

NORM NOTATION

Frobenius' element-wise norm:

$$||A_{m \times n}||_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2} = \sqrt{\sum_{i,j} a_{ij}^2}$$

Notation for error:

$$||X - Y||_F^2 = ||X - Y||^2 = \sum_{i,j} (x_{ij} - y_{ij})^2$$

NMF AS ERROR-MINIMIZATION

COMPUTATIONAL PROBLEM

Input: $V_{n \times m}$

Minimize $\|V - WH\|^2$

subject to $W, H \geq 0$.

- choose the new dimension r s.t. $(n + m)r < nm$;
- calculate $W_{n \times r}$ and $H_{r \times m}$.

ASIDE: INFORMATION-THEORETIC VIEW

If the input matrix can be (somehow) normalised then we see the search for the perfect non-negative decomposition in terms of minimizing *divergence*:

$$D_I(X||Y) = \sum_{i,j} (x_{ij} \cdot \log(\frac{x_{ij}}{y_{ij}}) - x_{ij} + y_{ij})$$

Minimize $D_I(V||WH)$

subject to $W, H \geq 0$.

Recommended version for sparse counting data.

The Kullback-Leibler divergence, D_{KL} , may also be used.

GRADIENT DESCENT MAY NOT FIND THE MINIMAL-ERROR SOLUTION

Although [error func.] are convex in W only or H only, they are not convex in both variables together.

Therefore it is unrealistic to expect an algorithm to solve [the problem] in the sense of finding global minima.

However, there are many techniques from numerical optimization for finding local minima.

Gradient descent is perhaps the simplest technique to implement, but convergence can be slow.

LEE-SEUNG'S METHOD

ITERATED ERROR BALANCING

1. initialise W and H at random

repeat:

2. compute the error

3. update W and H with the **multiplicative update rule**

4. normalise W : each column sums to 1

until error is tamed

$$\begin{aligned} W_{ia} &\leftarrow W_{ia} \sum_{\mu} \frac{V_{i\mu}}{(WH)_{i\mu}} H_{a\mu} \\ W_{ia} &\leftarrow \frac{W_{ia}}{\sum_j W_{ja}} \end{aligned} \quad \begin{aligned} H_{a\mu} &\leftarrow H_{a\mu} \sum_i W_{ia} \frac{V_{i\mu}}{(WH)_{i\mu}} \end{aligned}$$

MULTIPLICATIVE UPDATE

Classical Gradient descent: we *move around* by adding/subtracting some quantity

NMF: we *move around* by multiplying by a *local* error measure $\frac{v_{i\mu}}{(wh)_{i\mu}}$

$$\begin{aligned} W_{ia} &\leftarrow W_{ia} \sum_{\mu} \frac{V_{i\mu}}{(WH)_{i\mu}} H_{a\mu} \\ W_{ia} &\leftarrow \frac{W_{ia}}{\sum_j W_{ja}} \end{aligned} \quad \begin{aligned} H_{a\mu} &\leftarrow H_{a\mu} \sum_i W_{ia} \frac{V_{i\mu}}{(WH)_{i\mu}} \end{aligned}$$

$$W_{ia} \leftarrow W_{ia} \sum_{\mu} \frac{V_{i\mu}}{(WH)_{i\mu}} H_{a\mu}$$

$$W_{ia} \leftarrow \frac{W_{ia}}{\sum_j W_{ja}}$$

$$H_{a\mu} \leftarrow H_{a\mu} \sum_i W_{ia} \frac{V_{i\mu}}{(WH)_{i\mu}}$$

- through iteration, the $\frac{v_{i\mu}}{(wh)_{i\mu}}$ factors vanish and we stop.
- the update rules maintain non-negativity and force the $\mathbf{w}_i$ columns to sum to 1.

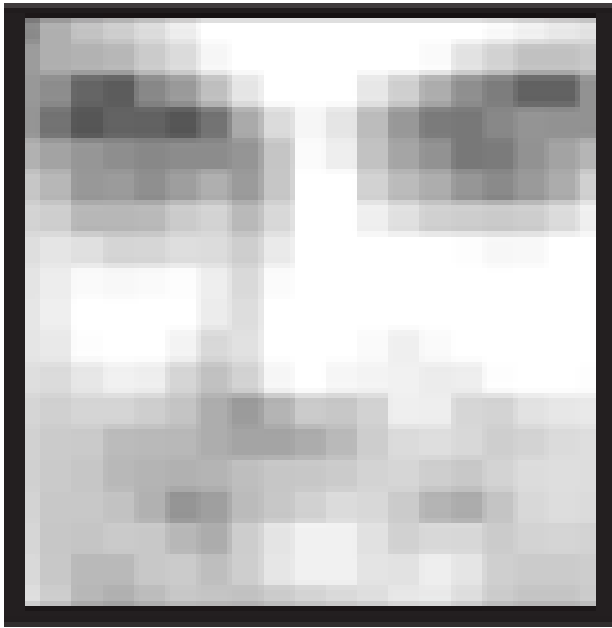
INTERPRETABILITY OF NMF

EXTRACTING COMMON PARTS: FACIAL FEATURES

Each column V_i represent a 19x19 mugshots

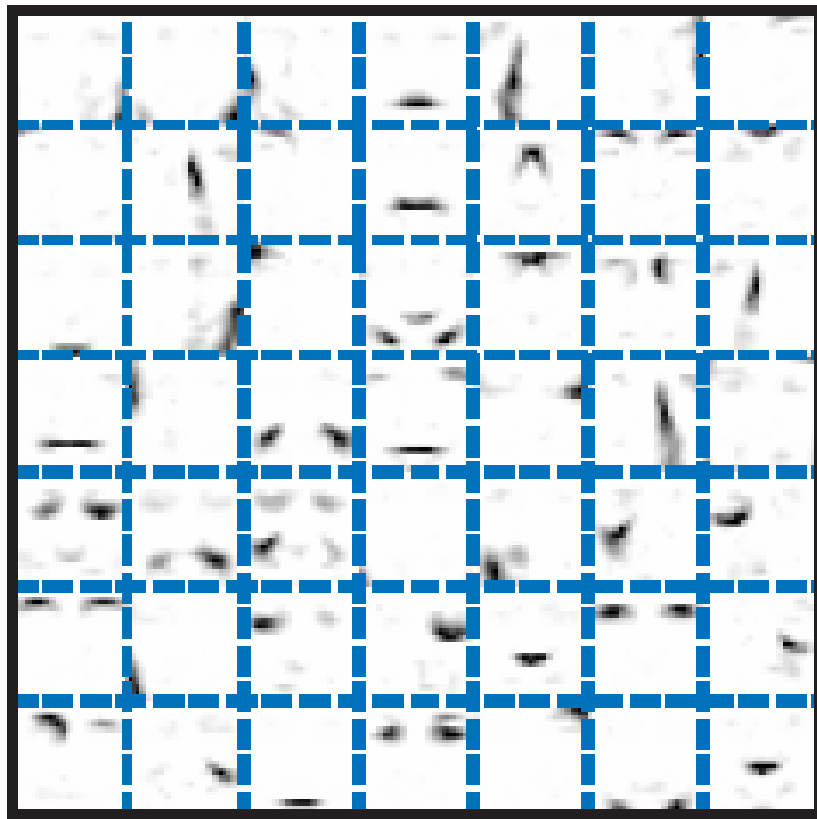
Lee-Seung render it as a linear combination of 49 common facial features, each weighted by an entry in the H matrix.

Original

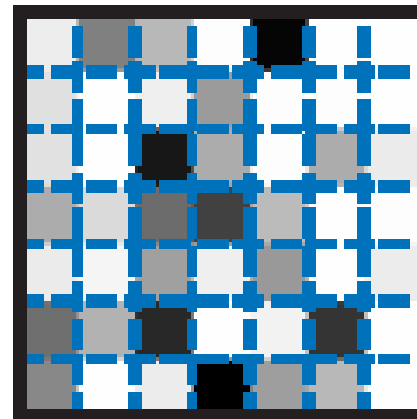


W AND H IN A 7X7 MONTAGE

NMF

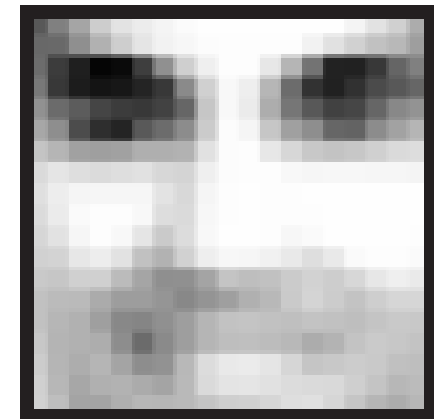
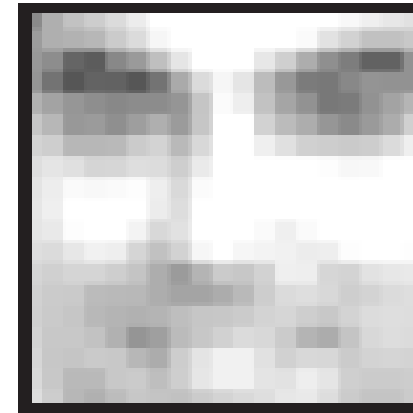


$\times$



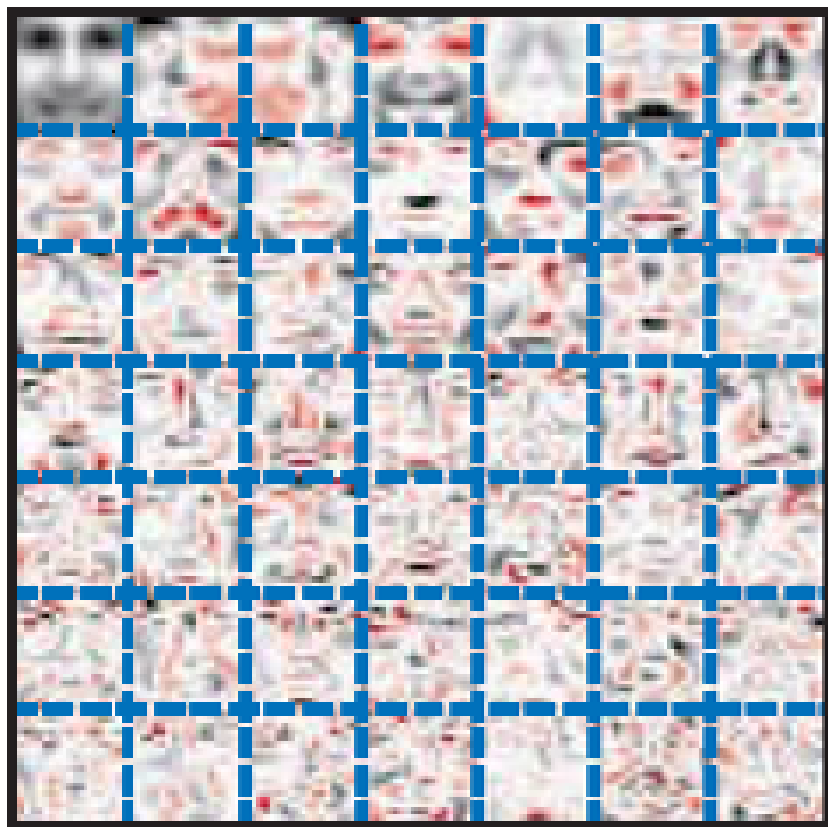
$=$

Original

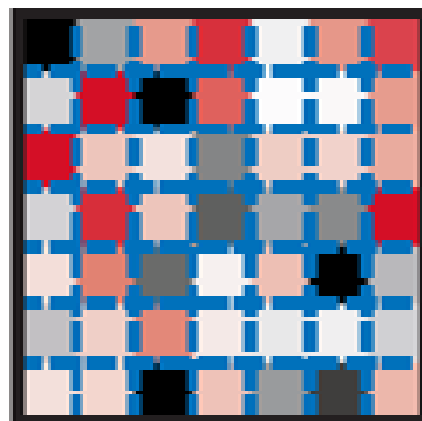


The *eigenfaces* might have negative values

PCA



$\times$



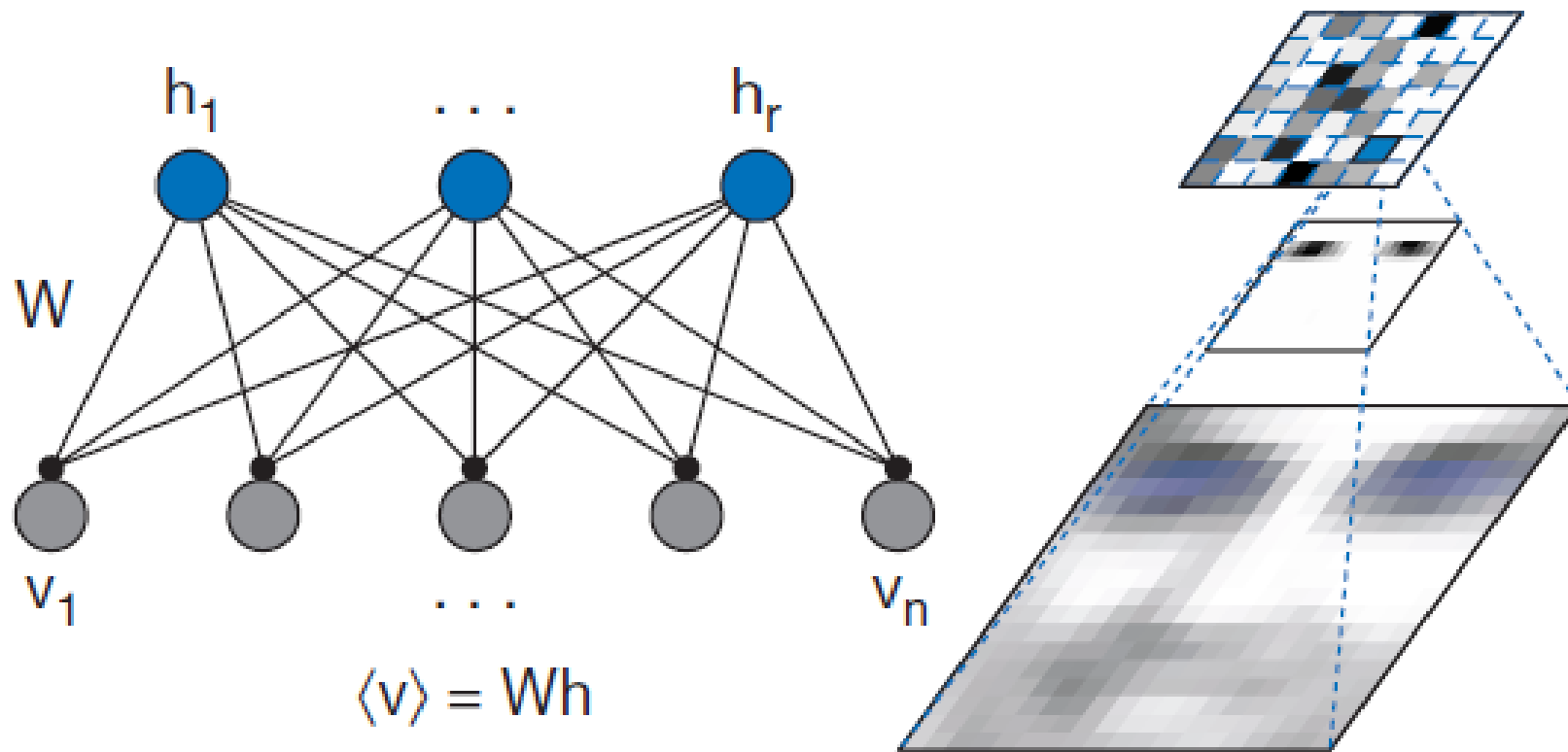
$=$



A PROBABILISTIC HIDDEN-VARIABLES MODEL:

The columns. of W are *bases* that are combined to form the reconstruction

As in neural networks, the influence of $\mathbf{h}_a$ on $\mathbf{v}_i$ is represented by a connection of strength w_{ia}



NMF IN SCIKIT-LEARN

THE OLIVETTI FACES

Today Lee-Seung method is implemented in Scikit-learn via the `mu` (multiplicative update) method

try `sklearn.decomposition.NMF`

```
1 faces = fetch_olivetti_faces()
2
3 nmf_estimator = decomposition.NMF(solver='mu')
4
5 nmf_estimator.fit(faces)
```

For reference: a [visual comparison](#) of the decomposition methods available in Scikit-learn

CODA: SIMPLE NMF EXAMPLE

DECOMPOSING THE NETFLIX EXAMPLE

Another look at the running example with film ratings

W: users' commitment to a topic.

H: films' pertinence to a specific topic (binary, why?)

	Matrix	Alien	Star Wars	Casablanca	Titanic
Joe	1	1	1	0	0
Jim	3	3	3	0	0
John	4	4	4	0	0
Jack	5	5	5	0	0
Jill	0	0	0	4	4
Jenny	0	0	0	5	5
Jane	0	0	0	2	2

Figure 11.6: Ratings of movies by users

Let R be the film ratings matrix and compute $R = PQ$
 $M = 7, N = 5$ and we fix $K = 2$ to run NMF:

```
1  @INPUT:
2      R: input matrix to be factorized, dim. M x N
3      P: an initial m. of dim. M x K
4      Q: an initial m. of dim. K x N
5      K: the no. of latent features
6      steps: the max no. of steps to perform the optimisation
7      alpha: the learning rate
8      beta: the regularization parameter
9
10 @OUTPUT:
11     the final matrices P and Q
```

DIRECT IMPLEMENTATION (1 RUN)

```
1 nP=  
2 [[ 0.33104196  0.39332058]  
3  [ 1.08079793  1.08397306]  
4  [ 1.59267325  1.27929568]  
5  [ 1.87852789  1.72209575]  
6  [ 0.67146598  1.76523621]  
7  [ 1.04872774  2.10824903]  
8  [ 0.94419145  0.59698619]]
```

```
1 nQ.T=  
2 [[ 1.27381876  1.3870236  1.67315614  0.9855609  0.81578369]  
3  [ 1.50953822  1.38352352  1.06501557  1.87281749  1.96189735]]
```

ANALYSIS OF THE ERROR, I

```
1 np.dot(nP, nQ.T) = [  
2     [ 1.01541991  1.00333129  0.97277743  1.06287968  1.04171324]  
3     [ 3.01303945  2.99879446  2.96279189  3.09527589  3.0083412 ]  
4     [ 3.95992279  3.97901104  4.02726085  3.9655638   3.80912366]  
5     [ 4.99247343  4.98812249  4.97712927  5.07657468  4.91104751]  
6     [ 3.52001748  3.37358497  3.00347147  3.96773585  4.01098322]  
7     [ 4.51837154  4.37142223  4.00000329  4.98195069  4.99170315]  
8     [ 2.10390225  2.13556026  2.21557931  2.04860435  1.94148161]  
9 ]
```

```
1 ratings = [[1, 1, 1, 0, 0],  
2            [3, 3, 3, 0, 0],  
3            [4, 4, 4, 0, 0],  
4            [5, 5, 5, 0, 0],  
5            [0, 0, 0, 4, 4],  
6            [0, 0, 0, 5, 5],  
7            [0, 0, 0, 2, 2]  
8            ]
```

ANALYSIS OF THE ERROR, II

```
1 np.rint(np.dot(nP, nQ.T))= [  
2     [ 1.  1.  1.  1.  1.]  
3         [ 3.  3.  3.  3.  3.]  
4         [ 4.  4.  4.  4.  4.]  
5         [ 5.  5.  5.  5.  5.]  
6         [ 4.  4.  4.  4.  4.]  
7         [ 5.  5.  5.  5.  5.]  
8         [ 2.  2.  2.  2.  2.]  
9     ]
```

```
1 ratings = [[1, 1, 1, 0, 0],  
2            [3, 3, 3, 0, 0],  
3            [4, 4, 4, 0, 0],  
4            [5, 5, 5, 0, 0],  
5            [0, 0, 0, 4, 4],  
6            [0, 0, 0, 5, 5],  
7            [0, 0, 0, 2, 2]]
```


ANALYSIS OF THE RESULT

```
1 W(user x topic) = [  
2   [ 0.          0.82037571]  
3   [ 0.          2.46112713]  
4   [ 0.          3.28150284]  
5   [ 0.          4.10187855]  
6   [ 1.62445593  0.          ]  
7   [ 2.03056992  0.          ]  
8   [ 0.81222797  0.          ]  
9   ]
```

```
1 H(topic x film) =  
2 [[ 0.          0.          0.          2.46236289  2.46236289]  
3  [ 1.21895369  1.21895369  1.21895369  0.          0.          ]]
```

W: users' commitment to a topic.

H: films' pertinence to a specific topic (binary, why?)